

Homework 4

1 Hyperbolic Graph Embedding

1. Hyperboloid model (aka Minkowski model or Lorentz model) is a model of n -dimensional hyperbolic geometry in which points are represented in the upper sheet of a two-sheeted Hyperboloid in $(n + 1)$ -dimensional Minkowski space as shown in Figure 1. Consider the 3D Minkowski space, 2D Hyperboloid satisfies the Cartesian coordinates equation $x^2 + y^2 - z^2 = -1$, where $z > 0$.

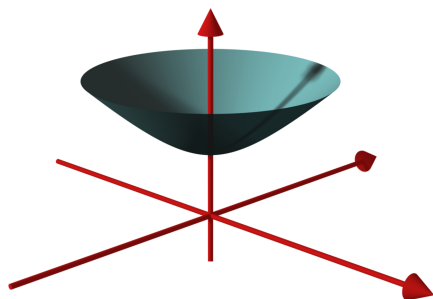


Figure 1: Hyperboloid Model

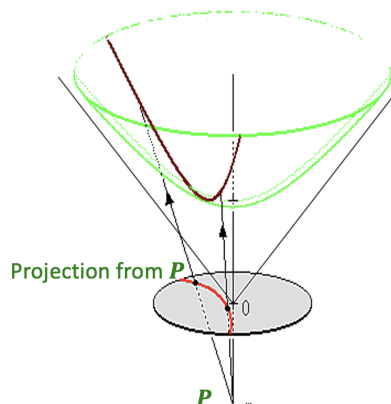


Figure 2: Projection of Hyperboloid Model to Poincaré Model

The Poincaré model can be obtained with a projection of the Hyperboloid model onto the $z = 0$ plane from a focal point P located on the z -axis, as shown in Figure 2.

- (a) Identify the coordinates of the focal point P from Figure 2, and derive expressions for the projected point (x^p, y^p) on the 2D Poincaré disk in terms of the Hyperboloid coordinates (x^h, y^h, z^h) .
- (b) Using the hyperboloid constraint $(x^h)^2 + (y^h)^2 - (z^h)^2 = -1$ with $z^h > 0$, prove that the projected point (x^p, y^p) always lies strictly inside the unit disk, i.e., $(x^p)^2 + (y^p)^2 < 1$.
- (c) Derive the inverse mapping: given a point (x^p, y^p) in the Poincaré disk with $(x^p)^2 + (y^p)^2 < 1$, express the corresponding Hyperboloid coordinates (x^h, y^h, z^h) in terms of x^p and y^p .

Hint: Let $r^2 = (x^p)^2 + (y^p)^2$ and use the result from part (b).

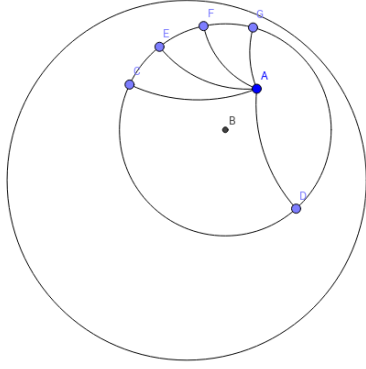


Figure 3: Hyperbolic Circle

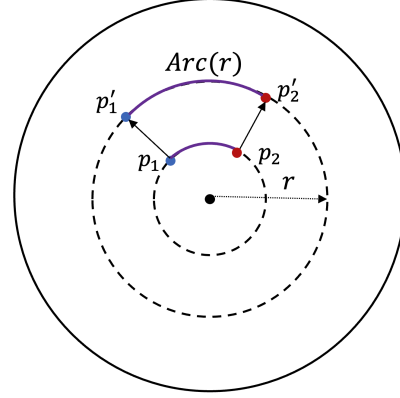


Figure 4: Poincaré Disk

2. Consider a hyperbolic circle in the 2D Poincaré model in which all points have a fixed distance from the center of the circle. The fixed distance is called the Hyperbolic radius r . Due to the negative curvature of the Poincaré model, the distance of points increases as they are closer to the boundary. Therefore, the center of the hyperbolic circle is different from the euclidean center. As shown in Figure 3, the center of the hyperbolic circle is A and all points on the circle (e.g., C, D, E, F, G) have the same distance from the center A .

The Poincaré disk metric in polar coordinates (ρ, θ) centered at the origin is:

$$ds^2 = \frac{4(d\rho^2 + \rho^2 d\theta^2)}{(1 - \rho^2)^2}$$

where $\rho \in [0, 1)$ is the Euclidean radial coordinate. The hyperbolic distance from the origin to a point at Euclidean radius ρ is $r = 2 \operatorname{arctanh}(\rho)$, equivalently $\rho = \tanh(r/2)$.

- (a) For a hyperbolic circle of radius r centered at the origin (i.e., the set of points at hyperbolic distance r from the origin), derive the circumference $C_h(r)$ by integrating ds along the circle. Then compare $C_h(r)$ with the Euclidean circumference $C_e(r) = 2\pi r$ and prove which one is larger for $r > 0$.
Hint: On the circle, $\rho = \tanh(r/2)$ is constant and θ varies from 0 to 2π .
- (b) Using the circumference formula you derived in part (a) and the result from Question 1(c), show that a hyperbolic circle of radius r centered at the origin in the Poincaré disk corresponds to a Euclidean circle of Euclidean radius $\rho_E = \tanh(r/2)$. Use this to express the ratio $C_h(r)/C_e^{\text{disk}}$ where $C_e^{\text{disk}} = 2\pi\rho_E$ is the Euclidean circumference of the same disk-circle, and show this ratio grows without bound as $r \rightarrow \infty$.
- (c) As Figure 4 shows, point p_1 moves to p'_1 and p_2 moves to p'_2 . The two points move away from the origin along the radial direction of the Poincaré disk as the circle radius increases. Using the circumference formula from part (a), prove that $\text{Arc}(r)$ between p'_1 and p'_2 increases exponentially with respect to r .

2 Discrete Denoising Diffusion Models for Graph Generation

Markov Chain. Define a Markov Chain of discrete random variables $\{X_t\}$, where $t = 0, 1, 2, \dots$ indicate timestep. Each X_t denotes the present state at timestep t in the Markov Chain. Let $x_t \in$

$\{1, 2, \dots, K\}$ be an instantiation of the random variable X_t from its state space (i.e. $\{1, 2, \dots, K\}$). Let $\mathbf{x}_t$ be the corresponding one-hot encoding of x_t . That is to say, $\mathbf{x}_t$ is a row vector where the i -th element is 1 while others are 0s when $x_t = i$.

The transition probability from X_{t-1} to X_t is given by the transition matrix $Q_t \in \mathbb{R}^{K \times K}$, where the entry at (i, j) is defined as

$$[Q_t]_{ij} := p(X_t = j | X_{t-1} = i) \quad (1)$$

, which denotes the transition probability from $X_{t-1} = i$ to $X_t = j$.

1. Using the one-hot encoding $\mathbf{x}_t$ and the transition matrix Q_t , derive the following expressions. Clearly justify each step.

- (a) $p(X_t | X_{t-1} = x_{t-1})$ as a row vector. (*What does left-multiplying Q_t by a one-hot vector select?*)
- (b) Define $\bar{Q}_t = Q_1 Q_2 \dots Q_t$. Express $p(X_t | X_0 = x_0)$ as a row vector and explain why this multi-step formula holds by induction on t .
- (c) $p(X_t = x_t | X_{t-1} = x_{t-1})$ as a scalar.
- (d) $p(X_t = x_t | X_{t-1})$ as a column vector, where the i -th element is the transition probability from state i at timestep $t - 1$ to x_t .

2. Write out the expression of $p(X_{t-1} | X_t = x_t, X_0 = x_0)$ in terms of all the known variables.

Hint 1: Use Bayes' rule.

Hint 2: You can use $\odot$ for element-wise multiplication.

3. Suppose $Q_t = (1 - \beta)I + \beta \mathbb{1}^T \mathbb{1} / K$, where $\beta \in (0, 1)$, I is the identity matrix, and $\mathbb{1}$ is an all-one row vector.

- (a) Find a closed-form expression for $\bar{Q}_t = Q_1 Q_2 \dots Q_t$ and prove it by induction.
Hint: Compute $\bar{Q}_1, \bar{Q}_2, \bar{Q}_3$ first to identify the pattern. You may find the identity $\mathbb{1}^T \mathbb{1} \cdot \mathbb{1}^T \mathbb{1} = K \cdot \mathbb{1}^T \mathbb{1}$ useful.
- (b) Using your result from (a), determine what distribution $\mathbf{x}_t$ follows as $t \rightarrow \infty$, and explain why this limiting distribution is independent of the initial state x_0 .

4. Let G_0 be the initial state of an undirected graph with N nodes. We send G_0 to the Markov chain as described before to obtain $G_1, G_2, \dots, G_t$, where G_t is the state of the graph at timestep t . $G_t = (\{X_{i,t}\}, \{E_{ij,t}\})$, $1 \leq i < j \leq N$, where $X_{i,t}$ denote the state of the i -th node at timestep t , and $E_{ij,t}$ denotes the edge between the i -th node and the j -th node at timestep t (note that i, j in this problem is the index of nodes rather than the index of states). There are K states for nodes and 2 states for edges (existence or non-existence).

To simplify the problem, we make the following *i.i.d assumption*: we assume the Markov process is conducted on each node and each edge independently and identically. All nodes share the same set of transition matrices $\{Q_t^X : t = 1, 2, \dots\}$ and all edges share the same set of transition matrices $\{Q_t^E : t = 1, 2, \dots\}$.

Suppose we want to reverse the process of the Markov chain, i.e. $p(G_{t-1} | G_t = g_t)$, where $g_t \in \{1, \dots, K\}^N \times \{1, 2\}^{\frac{N(N-1)}{2}}$ is a realization of G_t . Due to the *i.i.d assumption*, we have

$$p(G_{t-1} | G_t = g_t) = \prod_i p(X_{i,t-1} | X_{i,t} = x_{i,t}) \prod_{i < j} p(E_{ij,t-1} | E_{ij,t} = e_{ij,t}) \quad (2)$$

- (a) Using the law of total probability by conditioning on $X_{i,0}$, expand $p(X_{i,t-1}|X_{i,t} = x_{i,t})$ as a weighted sum. Identify the coefficient (weight) of each term in the sum and express it using the results from Questions 1 and 2.

Hint: $P(A|B) = \sum_i P(A|B, C_i)P(C_i|B)$. Which quantity from Question 2 gives you $P(A|B, C_i)$?

- (b) Using part (a), write out the full expression for $p(G_{t-1}|G_t = g_t)$ by expanding both the node and edge terms. Briefly explain which components of this expression must be learned by a neural network in order to generate graphs by iterative denoising from $t = T$ down to $t = 0$.